

**VACANCY-DRIVEN ELECTRONIC STRUCTURE
MODULATION IN MONOLAYER WS_{2-x} : A
FIRST-PRINCIPLES APPROACH**

A Thesis

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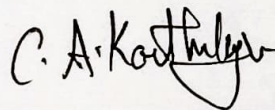
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APRIL 2026

DECLARATION

I, **KARTHIKEYAN C. A.**, hereby declare that the work entitled “**Vacancy-Driven Electronic Structure Modulation in Monolayer WS_{2-x} : A First-Principles Approach**” is an authentic record of work carried out under the supervision of **Dr. Rudra Banerjee**, Assistant Professor, Department of Physics and Nanotechnology, SRM Institute of Science and Technology, Kattankulathur.

The work presented in this thesis has not been submitted by me for the award of any other degree or diploma earlier.



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BONAFIDE CERTIFICATE

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ABSTRACT

Transition metal dichalcogenides (TMDs) have emerged as leading candidates for next-generation electronic, spintronic, and catalytic applications, owing to their stable two-dimensional hexagonal structure and highly tunable electronic and optical properties. Among TMDs, tungsten disulfide (WS_2) is of particular interest due to its sizeable bandgap and strong spin-orbit coupling. Point defects, especially chalcogen vacancies, occur naturally in 2D materials and can also be introduced deliberately through α -particle or electron-beam irradiation and thermal annealing, offering a practical route to tailor material properties and providing insight into the effects of substitutional doping.

In this work, we perform first-principles density functional theory calculations on a $4 \times 4 \times 1$ monolayer WS_2 supercell to systematically investigate the influence of sulfur vacancies on the electronic structure. The pristine monolayer yields a bandgap of 1.875 eV, in good agreement with previously reported values. Sulfur vacancy concentrations of 3%, 6%, 9%, and 12% are then introduced, and the resulting modifications to the band structure and density of states are analyzed. Our results elucidate the concentration-dependent evolution of vacancy-induced states within the gap, providing guidance for defect-engineered design of WS_2 -based devices.

Chapter 1

INTRODUCTION

The continuous expansion of the global semiconductor industry has created a strong demand for materials with tunable electronic, optical, and magnetic properties. Conventional bulk semiconductors, particularly silicon, have been the backbone of modern electronics for several decades; however, they are now approaching their fundamental and technological limits in terms of device miniaturization, energy efficiency, and multifunctionality. As devices continue to shrink and performance requirements increase, it becomes increasingly challenging for traditional materials to meet these demands [1]. Consequently, There is this growing need to search for an alternative material, that can offer greater flexibility and enhanced functionality for next-generation electronic and optoelectronic, spintronic and quantum applications. Within this context, 2D semiconductors are gaining significant attention as highly Favorable candidates owing to their reduced dimensionality and their unique physical characteristics, particularly TMDs.

A major motivation for investigating such materials comes from the fascinating phenomena of electronic phase transitions observed in a variety of condensed matter systems. In materials such as strongly correlated oxides and low-dimensional compounds, even subtle variations in external conditions—such as temperature, pressure, mechanical strain, or carrier concentration—can lead to dramatic changes in their electronic states. These transitions, which may occur between Semi-conducting, half-metallic, or even spin gapless semiconducting phases, are often accompanied by significant variations in electrical conductivity, optical and spintronic behavior, and magnetic ordering. The ability to control such transitions holds great potential for the development of advanced electronic devices, particularly in areas such as memory storage, sensing, and adaptive electronics.

Ever since the extraction of Graphene from graphite, using Mechanical exfoliation in 2004 [2], 2-D material has gathered significant attention. Two-dimensional crystals such as graphene and other TMDs exhibit electronic structures that strongly depend on their thickness and stacking configuration. As the number of layers increases, their

electronic properties undergo significant changes [3]. For instance, bulk MoS₂ exhibits an indirect-bandgap about 1.29 eV, whereas a monolayers with a thickness of about 0.65 nm shows a direct-bandgap of around 1.8 eV. This Transition from an indirect-bandgap to a direct-bandgap is attributed to the **Quantum Confinement Effect** [4, 5]. As a result, monolayer MoS₂ demonstrates high photoresponsivity, reaching values up to 880 A · W⁻¹, making it highly suitable for applications such as phototransistors and UV detectors due to its strong photoluminescence and optical absorption [6].

1.1 Transition Metal Dichalcogenides (TMDs)

Two-dimensional semiconducting Transition Metal Dichalcogenides, commonly known as TMDs, have the general molecular formula of MX₂, in which M refers to a Transition metal atom (namely Titanium, Molybdenum, Tungsten and Niobium) and X refers to chalcogen elements (namely Selenium, Sulfur and Tellurium). Such materials have attracted Considerable attention due to their diverse optical, electrical, spintronic and mechanical properties.

In the case of the bulk MX₂, individual layers are bound together by relatively weak van-der-Waals forces, whereas strong covalent bonds forms between metal and chalcogen atoms within each layer. Although the interlayer interaction is relatively weak, increasing the number of layers introduces collective effects such as interlayer coupling, geometrical confinement, and symmetry variations. These factors significantly influence the electronic structure, often resulting in an indirect bandgap in bulk systems [3]. In this bulk form, Tungsten Disulfide (WS₂) and MoS₂ exhibit nearly identical lattice constants (3.16 and 3.15 Å, respectively). A key distinction between MoX₂ and WX₂ systems lies in the strength of spin–orbit coupling, which is significantly larger in tungsten-based compounds (approximately 420 meV) due to the heavier atomic mass of tungsten [7]. Consequently, WS₂ exhibits stronger spin–orbit effects and enhanced photoluminescence compared to MoS₂ [6].

Various synthesis techniques have been developed to fabricate TMD materials with controlled thickness and high quality. Among these, mechanical exfoliation represents one of the earliest approaches employed to isolate monolayer TMDs from bulk crystals. Although this method yields high-quality samples, it is not suitable for large-scale production. Alternative approaches, namely Liquid-phase Exfoliation, Chemical-Vapor-Deposition (CVD), and Hydrothermal Synthesis, have been extensively investigated to enable scalable and uniform growth of TMD films. [8].

By introducing point defects or through substitutional doping, it is possible to modify key properties such as the band gap, carrier concentration, and transport characteristics, and in some cases even induce magnetic behavior. This level of control makes TMDs highly versatile materials for nanoscale device engineering, where precise manipulation of

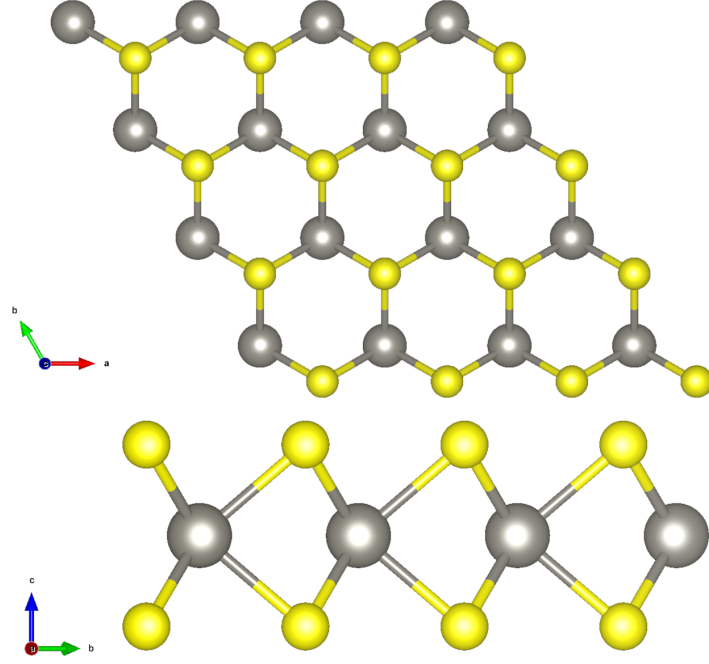


Figure 1.1: Top and side views of the optimized $4 \times 4 \times 1$ single-layer WS_2 supercell are shown. Gray spheres represent tungsten (W) atoms, while yellow spheres represent sulfur (S) atoms.

electronic properties is essential.

1.2 Tungsten Disulfide (WS_2)

Monolayer WS_2 demonstrates superior optical properties, including a direct bandgap, enhanced optical response, and higher photoluminescence quantum yield, which are not observed in its bulk counterpart (approximately 6%) [9]. Structurally, monolayer WS_2 possesses a hexagonal lattice with space group $P6_3/mmc$ and exhibits three-fold rotational symmetry (C_3 symmetry in the basal plane). This symmetry, combined with strong W–S covalent bonding, contributes to its high mechanical and thermal stability. The electronic properties of WS_2 are strongly dependent on thickness. A monolayer exhibits a direct-bandgap about 2 eV, whereas bandgap decreases to about 1.2 eV and becomes indirect in bulk form. More specifically, for thicknesses of [30, 12, 6, 1] nm, the corresponding bandgaps are [1.50, 1.63, 1.72, 1.79] eV [10] [11]. Naturally occurring bulk WS_2 is found as the mineral tungstenite, which also possesses a hexagonal structure and an indirect bandgap. Due to its layered structure and weak interlayer interactions, it is widely used as a solid lubricant. Under laboratory conditions, high-quality 2H- WS_2 are synthesized via techniques such as CVT (Chemical Vapor Transport), CVD (Chemical Vapor Deposition), and ALD (Atomic Layer Deposition) [7, 9, 12, 13].

Given its unique electronic and optical properties, WS_2 has a wide range of applications. In photocatalytic water splitting, materials with bandgaps of approximately

1.5–2.8 eV are desirable to achieve solar-to-hydrogen efficiencies of around 10% for practical viability. WS_2 -based nanotubes have been demonstrated as efficient sunlight-driven water-splitting catalysts. Furthermore, the Incorporation of dopants or structural defects can significantly enhance photocatalytic performance by increasing light absorption coefficients [14]. Additionally, WS_2 is considered a promising hydro-treating catalyst used in petroleum refining processes. Although Mo-based and W-based catalysts share similar structural characteristics, they differ in properties such as oxide-to-sulfide conversion, hydrogenation activity, and hydrodesulfurization efficiency [15]. Owing to its high carrier mobility and reduced effective mass, WS_2 is also widely utilized in field-effect transistors (FETs), including single-layer and bilayer semiconductor devices [16].

Furthermore, WS_2 finds applications as an electrode material [17]. In biomedical applications, materials such as WS_2 and WSe_2 have been explored as drug delivery carriers for tuberculosis treatment, as their bandgap remains largely unaltered upon adsorption of drugs such as Pyrazinamide (PZA) and Isoniazid (INH). Drug release can be triggered using near-infrared or visible light [18]. Beyond these applications, WS_2 also holds significant potential in emerging fields such as spintronics and quantum technologies.

1.3 Defects

Defects are inherently formed in two-dimensional (2D) materials, even under controlled laboratory conditions during sample growth. Therefore, these defects must be carefully examined and understood. They strongly influence electronic, optical and magnetic properties of TMDs. Owing to their 2D nature and reduced dielectric screening, TMDs are highly sensitive to local perturbations, making point defects a significant factor in tuning their physical properties. [19, 20].

Among various types of point defects, chalcogen vacancies (X vacancies) are the most commonly observed intrinsic defects in TMDs. They are not only readily formed but also significantly alter the electronic structure and can induce magnetism, making them promising for spintronic applications [21]. These vacancies typically originate during synthesis processes such as chemical vapor deposition or exfoliation, and can also be intentionally introduced through α -particle or electron beam irradiation, as well as thermal annealing. Chalcogen vacancies give rise to localized electronic states within the band gap, which can substantially modify the electronic structure by narrowing the band gap or enabling defect-mediated conductivity. In many cases, these defect states predominantly arise from the Transition metal atom’s d -orbitals adjacent to the vacancy.

In addition to vacancies, substitutional defects—where an atom is replaced by a different element—provide another effective means of tuning material properties. Doping with

foreign atoms can introduce donor or acceptor levels, modify carrier concentration, and in some cases induce spin polarization. The nature of the defect states depends strongly on the electronic configuration of the dopant and its interaction with the host lattice.

Overall, point defects in TMDs serve as a powerful tool for engineering material properties, enabling control over conductivity, optical response, and magnetic behavior. Understanding the formation, stability, and electronic signature of these defects is essential for the development of TMD-based nanoelectronic and spintronic devices.

This thesis is structured into four chapters. Chapter 1 presents the motivation for the study along with a comprehensive review of the literature, Concerning structural and electronic properties of transition metal dichalcogenides, with particular emphasis on WS_2 . It also discusses point defects, especially vacancy defects, in two-dimensional WS_2 , along with their influence on material properties. In addition, the fundamental concepts of Density-Functional-Theory (DFT) and the Hubbard-Correction (DFT+U) are introduced. Chapter 2 outlines the computational methodology employed in this work, including the construction of pristine and defected supercells, the incorporation of sulfur vacancies, oxygen doping, and the computational parameters used in the simulations. Chapter 3 presents the results of structural optimization and electronic structure calculations, followed by a detailed analysis and discussion in the context of existing literature. Finally, Chapter 4 summarizes the key findings of the study and highlights possible directions for future research.

Chapter 2

METHODOLOGY

Density Functional Theory (DFT) is among one of the most commonly employed theoretical frameworks for investigating the electronic properties of materials. Several computational approaches are available for this purpose, including the Bethe–Salpeter equation (BSE), the Hartree–Fock method, Quantum Monte Carlo simulations, Green’s function methods, the GW approximation, and machine learning techniques. Among these, DFT is particularly prominent Because of its advantageous balance between computational accuracy and efficiency.

DFT is a first-principles approach in which the complex many-body problem of interacting electrons is transformed into an equivalent system of non-interacting particles. This mapping is realized through the Kohn–Sham equations [22], which are Schrödinger-like equations derived from the many-body Hamiltonian in a reduced form, given by

$$\hat{H} = - \sum_I \frac{\hbar^2}{2M} \nabla_I^2 - \sum_i \frac{\hbar^2}{2m} \nabla_i^2 + \sum_{I \neq J} \frac{Z^2 e^2}{|R_I - R_J|} + \sum_{i \neq j} \frac{e^2}{|r_i - r_j|} - \sum_{I \neq i} \frac{Z e^2}{|R_I - r_i|}. \quad (2.1)$$

The leading two terms correspond to the Kinetic-energies of nuclei and electrons, respectively. Remaining three terms accounts for the Coulomb interactions within the system. Specifically, the third term represents the nucleus–nucleus repulsion, The fourth term models the electron–electron interaction, and the final term corresponds to the electron–nucleus interaction. Here, "M" and "m" denote the masses of the nuclei and the electron, respectively, while " R_I (R_J)" and " r_i (r_j)" represent the positions of the nuclei and electrons. The quantity e denotes the elementary charge, and " Z " denotes the atomic number (i.e., number of protons in the nucleus). " $\hbar$ " represents the reduced Planck constant.

Hamiltonian in Eq. 2.1 contains both electronic and nuclear coordinates. The resulting many-body problem cannot be solved analytically; therefore, a series of approximations and numerical methods must be employed. The first step in this process is the Born–

Oppenheimer approximation (Eq. 2.2), this assumes that, the dynamics of electrons and nuclei can be treated independently due to their large mass difference. Owing to this disparity, It is assumed that the electrons adjust instantaneously to the positions of the nuclei, which are treated as effectively stationary. Consequently, the total wavefunction describing the system, $\Psi(R, r)$ can be separated into electronic and nuclear components as

$$\hat{\Psi}(R, r) = \hat{\psi}(r) \hat{\phi}(R), \quad (2.2)$$

In which the electronic wave-function is represented by $\psi(r)$ and nuclear wavefunction is denoted by $\phi(R)$. Within this framework, the many-body Hamiltonian can be effectively decoupled, allowing the electronic and nuclear parts to be solved independently.

In the full many-body Hamiltonian, the electron–electron interaction is a two-body term. This term makes the problem computationally intractable for large systems due to its explicit dependence on electron pairs. In the Hartree approximation, this many-body interaction is simplified by introducing electron density $\rho(r)$, defined as

$$\rho(r) = \sum_{i=1}^N |\hat{\psi}_i(r)|^2. \quad (2.3)$$

Consequently, the discrete sum over electron pairs is replaced by a continuous charge distribution. Accordingly, the Coulombic electron–electron interaction can be represented as that of a single electron interacting with the average charge density:

$$V_H(r) = e^2 \int \frac{\rho(r')}{|r - r'|} dr'. \quad (2.4)$$

Thus, the two-body interaction term is transformed into an effective one-body potential, known as the Hartree potential. This replaces the summation over all electron pairs with an integral over the charge density, significantly simplifying the problem. With this, the Hamiltonian can be rewritten as

$$\hat{H}_{\text{Hartree}} = \sum_i \left[-\frac{\hbar^2}{2m} \nabla_i^2 - \sum_I \frac{Ze^2}{|R_I - r_i|} + V_H(r_i) \right]. \quad (2.5)$$

The Hartree method neglects the spin of electrons and fails to enforce the antisymmetry of the electronic wavefunction, i.e., $\psi(q_i, q_j) = -\psi(q_j, q_i)$. To address this, the wavefunction is often approximated using a Slater determinant, which inherently satisfies the antisymmetry requirement and naturally incorporates exchange interactions. This leads to the Hartree-Fock formalism, in which the Hamiltonian is expressed in terms of the Hartree-Fock operator and single-electron orbitals:

$$\left[\hat{h}(i) + \sum_j^{occ} \left(\hat{J}_j(i) - \hat{K}_j(i) \right) \right] \hat{\psi}_i = \epsilon_i \hat{\psi}_i \quad (2.6)$$

where $\hat{K}_j(i)$ and $\hat{J}_j(i)$ denote the Exchange and Coulomb operators, respectively, and $\hat{h}(i)$ represents the single-electron Hamiltonian, which includes both the Kinetic energy of i^{th} electron and its potential energy arising from the nuclei. And ψ_i is single electron orbital.

The Hartree-Fock method, while accounting exactly for the exchange interaction through the Slater determinant, neglects dynamic electron correlation, which arises from the instantaneous interactions between electrons. As a result, it systematically overestimates the ground-state energy and cannot capture subtle many-body effects, such as van der Waals interactions or strongly correlated electronic states. Furthermore, the computational cost of Hartree-Fock scales poorly with system size, making it challenging to apply to large or complex systems. These limitations highlight the need for a more accurate and computationally efficient framework. The theoretical foundation for such a framework is provided by the **Hohenberg–Kohn theorems** [23]:

1. **Theorem 1:** *Ground-state electron density, $\rho_0(\mathbf{r})$, uniquely determines external potential $V_{\text{ext}}(\mathbf{r})$, up to an additive constant. Consequently, all ground-state properties can be formulated as functionals of $\rho_0(\mathbf{r})$.*
2. **Theorem 2:** *There exists a universal functional $F[\rho]$, which is independent of $V_{\text{ext}}(\mathbf{r})$, allowing the total energy of the system to be expressed in terms of electron density $\rho(\mathbf{r})$.*

$$E[\rho] = F[\rho] + \int \rho(r) V_{\text{ext}}(r) dr \quad (2.7)$$

is minimized by the true ground-state density $\rho_0(r)$. The Universal functional can be defined as

$$F[\rho] = T[\rho] + V_{ee}[\rho], \quad (2.8)$$

in which $T[\rho]$ represents Kinetic energy of the electrons, and $V_{ee}[\rho]$ is the energy functional of exact electron–electron interaction.

These theorems provide the theoretical foundation for representing the interacting many-electron system by an auxiliary system of non-interacting electrons, which exactly reproduces the ground-state electron density and gives rise to the *Kohn–Sham equations*:

$$\left[-\frac{1}{2} \nabla^2 + V_{\text{ext}}(r) + \int \frac{\rho(r')}{|r - r'|} dr' + \frac{\delta E_{\text{xc}}[\rho]}{\delta \rho(r)} \right] \hat{\psi}_i(r) = \epsilon_i \hat{\psi}_i(r) \quad (2.9)$$

where $V_{\text{ext}}(r)$ represents external potential, $\rho(r)$ denotes electronic charge density and $E_{\text{xc}}[\rho]$ is the exchange and correlation functional.

The total energy functional of the system is given by,

$$E[\rho] = T[\{\hat{\psi}_i(\rho)\}] + E_{\text{ext}}[\rho] + E_H[\rho] + E_{\text{xc}}[\rho], \quad (2.10)$$

in which $T[\{\psi_i(\rho)\}]$ represents kinetic energy of the non-interacting electrons, $E_{\text{ext}}[\rho]$ represents electron–nuclei interaction, $E_H[\rho]$ denotes the Hartree energy describing classical Coulombic, electron–electron repulsion, and $E_{\text{xc}}[\rho]$ accounts for all exchange and correlation effects beyond the classical description [24].

In practical calculations, Kohn–Sham equations are iteratively solved within a self-consistent field (SCF) scheme. Due to a fact that, effective potential relies on the electron density, subsequently determined from the solutions of Kohn–Sham equation, resulting in a circular dependence.

The procedure begins with an initial guess of electron density $\rho(\mathbf{r})$, based on which an effective potential is generated. Kohn–Sham equations are then solved iteratively in order to obtain a new sets of wavefunctions $\psi_i(r)$, which are subsequently used to compute an updated electron density. This process is iteratively repeated until convergence is achieved, i.e., when the difference between successive electron densities falls below a predefined threshold (refer Figure.2.1).

The computational implementation of DFT requires the choice of a suitable basis set to represent the wavefunctions. Common approaches include plane-wave basis sets, which are widely used for periodic systems, and localized basis sets such as atomic orbitals for molecular systems. Additionally, pseudopotentials or Projector-Augmented-Wave (PAW) methods are often employed to simplify the treatment of core electrons, thereby reducing computational cost.

Another critical aspect in DFT is the approximations applied to the Exchange Correlation functionals. Most common approximations include the Local Density Approximation (LDA) and the Generalized Gradient Approximation (GGA), which differ in how they incorporate electron density variations. The selection of Exchange-Correlation functional strongly affects the accuracy of computed properties, including the band structure, total energy, and optical response.

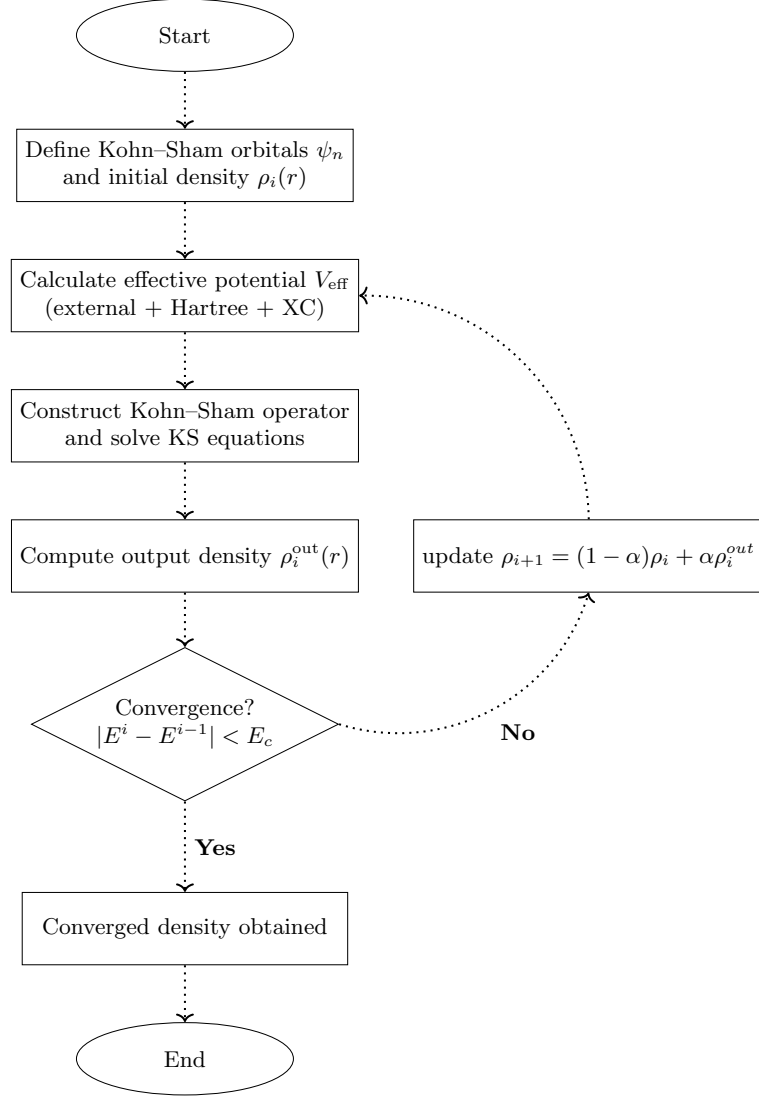


Figure 2.1: Schematic representation of the self-consistent field (SCF) loop used in density functional theory (DFT) calculations. The flowchart illustrates the iterative procedure for solving the Kohn-Sham equations.

Transition metals namely Tungsten, Molybdenum, Vanadium, and Titanium in transition metal dichalcogenides (TMDs) have highly localized d -orbitals. Electrons in these orbitals exhibit strong electron-electron correlations, which give rise to phenomena such as magnetism and superconductivity. Because of these strong correlations, the electrons cannot be accurately treated as independent particles, as assumed in standard Kohn-Sham Density Functional Theory (DFT). Consequently, standard DFT calculations may yield inaccurate results for TMDs. To improve the description of strongly correlated d electrons, we employ a Hubbard correction [25], known as DFT+ U , which adds an on-site Coulomb interaction term to the DFT energy:

$$E_{\text{DFT}+U} = E_{\text{DFT}} + \sum_a \frac{U_{\text{eff}}}{2} \text{Tr}(\rho^a - \rho^a \rho^a), \quad (2.11)$$

Here ρ^a represents occupation matrix of localized orbitals on atom a , and U_{eff} is the effective on-site Coulomb repulsion. While DFT+ U improves the treatment of localized electrons, it does not account for dynamical correlations, so more advanced methods may be required for highly correlated systems. In this study, the inclusion of U is essential to accurately capture the electronic structure and magnetic behavior of the transition metal d orbitals in TMDs [26].

First principles calculations were carried out within framework of DFT in order to investigate the electronic properties of pristine, S-vacancy induced, Oxygen doped monolayer. A monolayer geometry of $4 \times 4 \times 1$ pristine supercell consists of 48 atoms, including 16 tungsten atoms and 32 sulfur atoms, forming a layered structure in which, a layer of tungsten atoms is sandwiched between two layers of sulfur atoms. To prevent interactions between layers vacuum of 20 Å is incorporated along the $\vec{c}$ direction. The Generalized-Gradient-Approximation (GGA) in the Perdew–Burke–Ernzerhof (PBE) form was employed for the Exchange Correlation functional, along with Projector-Augmented-Wave (PAW) method. to properly address localized nature of d-electrons of Tungsten atoms, DFT+ U approach was empowered. The Hubbard U parameter of 2.87 eV was applied for d-orbit of Tungsten atoms as mentioned in literature [27]. Point defects in the form of sulfur vacancies (V_S) were introduced systematically into the $4 \times 4 \times 1$ supercell. Vacancy concentrations of 3%, 6%, 9%, and 12% were generated by removing 1, 2, 3, and 4 sulfur atoms, respectively (see Fig. 2.2). Plane-wave cutoff energy of 520 eV is used. Brillouin zone of was sampled using Monkhorst-Pack k-point of $1 \times 1 \times 1$ for structure optimization and $9 \times 9 \times 1$ was used for Electronic structure calculation.

The structural relaxation was performed with the lattice parameters fixed, allowing only the atomic positions to relax until forces on all atoms fell below convergence threshold. 0.01 eV/Å, and total energy converges within 10^{-5} eV. Spin polarised calculation were performed to account for all magnetic effects induced by Sulfur vacancy (V_S) and doping of Oxygen (V_S^O). The electronic structure of the material was analyzed with the spin resolved Density of States, including Total DOSs and PDOS of each atom to examine defect or doping induced polarization. The results were validated by comparison with available theoretical and experimental data.

The introduction of sulfur vacancies and subsequent oxygen substitution is expected to modify the electronic structure by introducing or suppressing localized defect states, altering the band structure, and shifting the Fermi level (E_F).

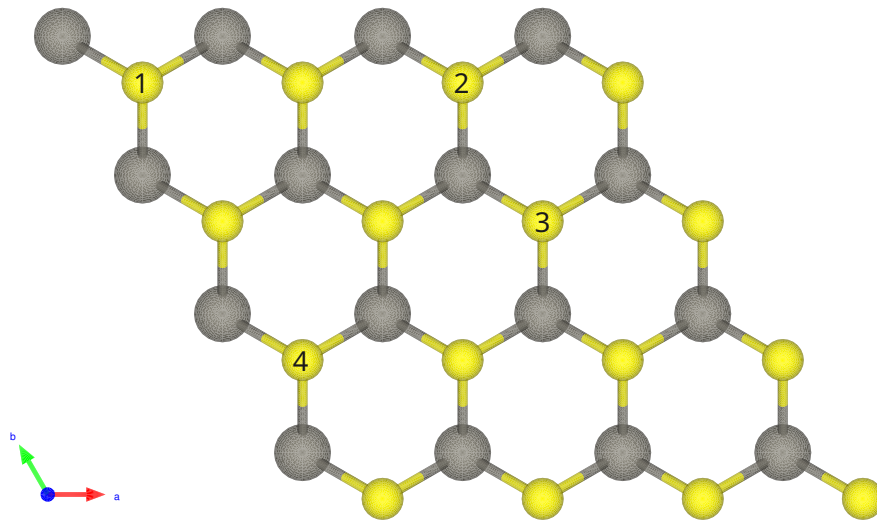


Figure 2.2: Top views of the single-layer WS₂ supercell, showing the sites where sulfur atoms have been removed and replaced by oxygen dopants, are marked 1–4 in the figure.

Chapter 3

RESULTS AND DISCUSSIONS

The Structure optimization calculation on the WS₂ Unit cell is performed and the optimized, relaxed structure is obtained. the results yields the lattice parameters to be $a = b = 3.1719 \text{ \AA}$, $c = 29.0953 \text{ \AA}$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$ and the bond length between Tungsten atoms and Sulfur atoms is $2.4151 \text{ \AA}$, Which is a good argument corresponding to the previously published experimental and theoretical results [28–31].

The Density Of States (DOS) quantifies the number of available electronic states per unit energy at each energy level, which is a crucial tool for analyzing material's electronic properties. Fermi-level is shifted to zero (i.e. $E_{new} = E_{old} - E_f$), for the convenience of observation. The Total and Partial Density of States of pristine system is given in the Figure 3.1. From the electronic calculations of the pristine super-cell, We infer that the 2d WS₂ monolayer material exhibits a semiconductor behavior, due to the absence of states on and above Fermi-level. We also found the bandgap of the semiconductor to be 1.861 eV which is close to the reference Theoretical 1.86 eV [32] and to the experimental results of 2.0 eV [33]. The symmetry between two spin channels suggest the non-magnetic behavior of the material. The calculated DOS is consistent with previous reports [34].

The Projected-Density of States illustrates that the valence-band is primarily composed of S-p orbitals with noticeable hybridization from W-d states, indicating strong covalent bonding between tungsten and sulfur atoms. In contrast, the conduction-band is dominated by W-d orbitals, suggesting that low-lying excited states are mainly of tungsten d-character. The absence of states around Fermi-level and the lack of mid-gap states indicate a defect-free and electronically stable system.

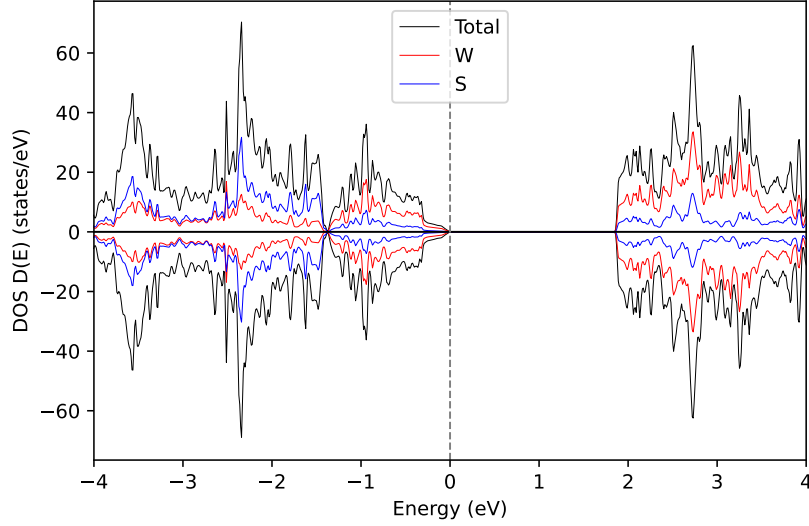


Figure 3.1: The total and partial density of states (DOS and PDOS) for the pristine WS_2 system are presented. In each plot, the dotted line denotes the Fermi level.

The spin-polarized density of states (DOS) of WS_2 with varying sulfur vacancy concentrations (3–12%) reveals that the system remains non-magnetic across all configurations. Spin-up channels and spin-down channels are symmetric, indicating the absence of exchange splitting and thus no spontaneous magnetic ordering. The introduction of sulfur vacancies leads to the emergence of defect-induced states near the Fermi-level, which become increasingly prominent with higher vacancy concentrations. Projected DOS analysis shows that these states are primarily derived from W-d orbitals, while S-p contributions remain relatively minor near the Fermi-level.

As the vacancy concentration increases, the band gap progressively decreases because of the emergence of defect-induced states near the Fermi-level, which become more prominent at higher concentrations (9% and 12%). Despite the presence of unsaturated bonds, which are often associated with vacancy-induced magnetism, no spin polarization is observed. This indicates that the defect states are sufficiently delocalized as a result of strong hybridization of W-d orbitals, preventing the formation of localized magnetic moments. Therefore, sulfur vacancies in WS_2 do not induce magnetism but instead introduce defect states that modify the electronic structure of the system.

Spin-polarized DOS for oxygen-doped WS_2 , where oxygen substitutes sulfur vacancy sites, reveals a non-magnetic ground state across all studied concentrations (3–12%). Spin-up channels and spin-down channels remain symmetric, indicating absence of exchange splitting and confirming that oxygen doping does not induce magnetism in the system. The results match with the previously published work [35]

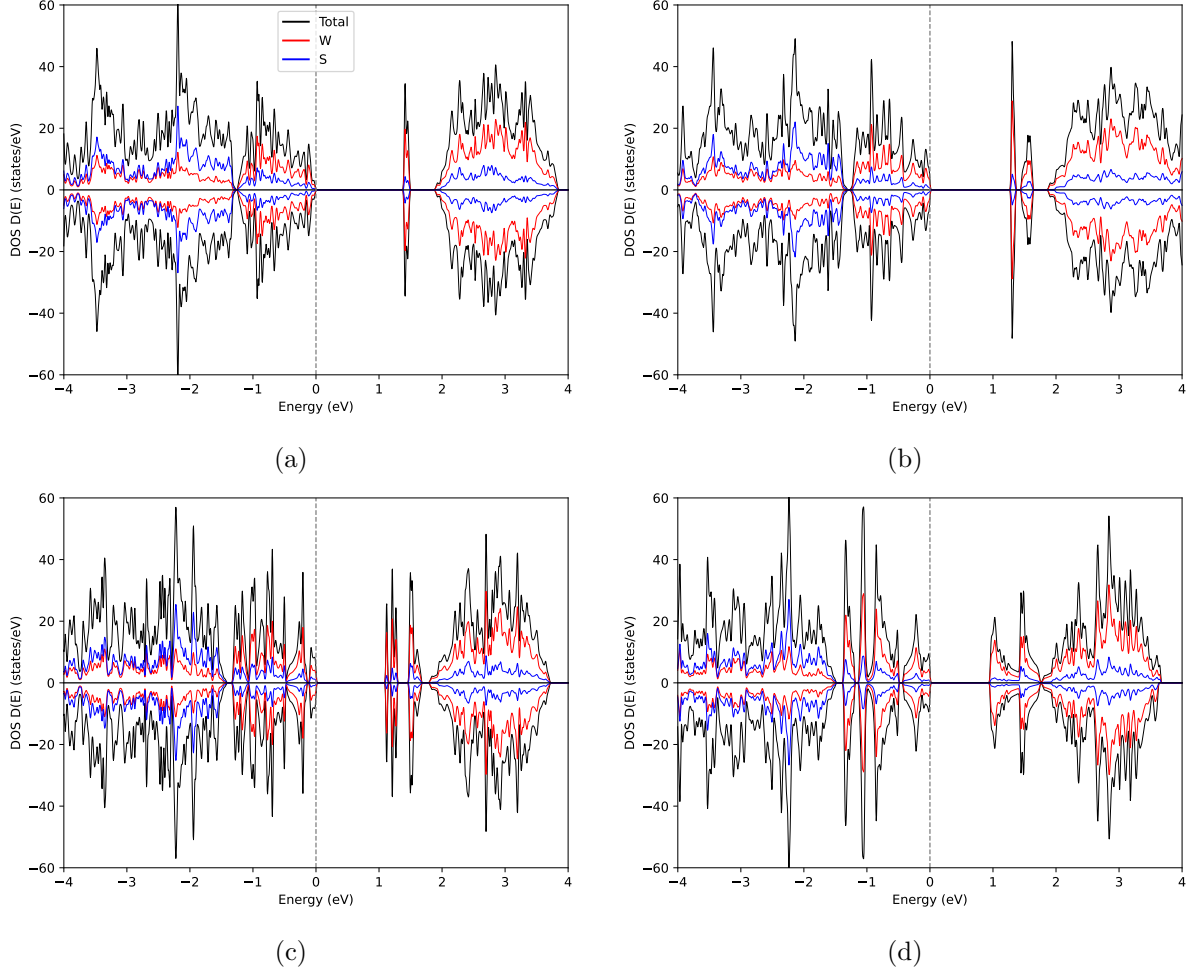


Figure 3.2: The total and Partial density of states (DOS and PDOS) plots are shown for vacancy-induced WS_2 systems with (a) 3%, (b) 6%, (c) 9%, and (d) 12% vacancy concentrations. In each plot, the dotted line indicates the Fermi level.

In contrast to the sulfur vacancy case, the introduction of oxygen significantly suppresses the defect-induced mid-gap states near the Fermi-level. This behavior indicates effective passivation of dangling bonds associated with sulfur vacancies. The projected DOS shows that while the electronic states near the Fermi-level are still primarily dominated by W-d orbitals, the overall intensity is reduced due to hybridization with O-p states. Oxygen contributions are mainly observed in the valence region, suggesting the formation of stable bonding states rather than localized defect states.

As the oxygen concentration increases, the system maintains a semiconducting character with no evidence of transition, unlike the sulfur-vacancy system. This indicates that oxygen plays a crucial role in restoring electronic stability by saturating the unsatisfied bonds of tungsten atoms. The absence of spin polarization further suggests that the introduced states are fully occupied and do not lead to localized magnetic moments. yet the calculations yield the net magnetic moment of $0.007 \mu_B$, nearly non magnetic state. Overall, oxygen substitution at sulfur vacancy sites acts as a defect passivator,

eliminating mid-gap states and preventing the emergence of magnetism, while stabilizing the electronic structure of WS₂.

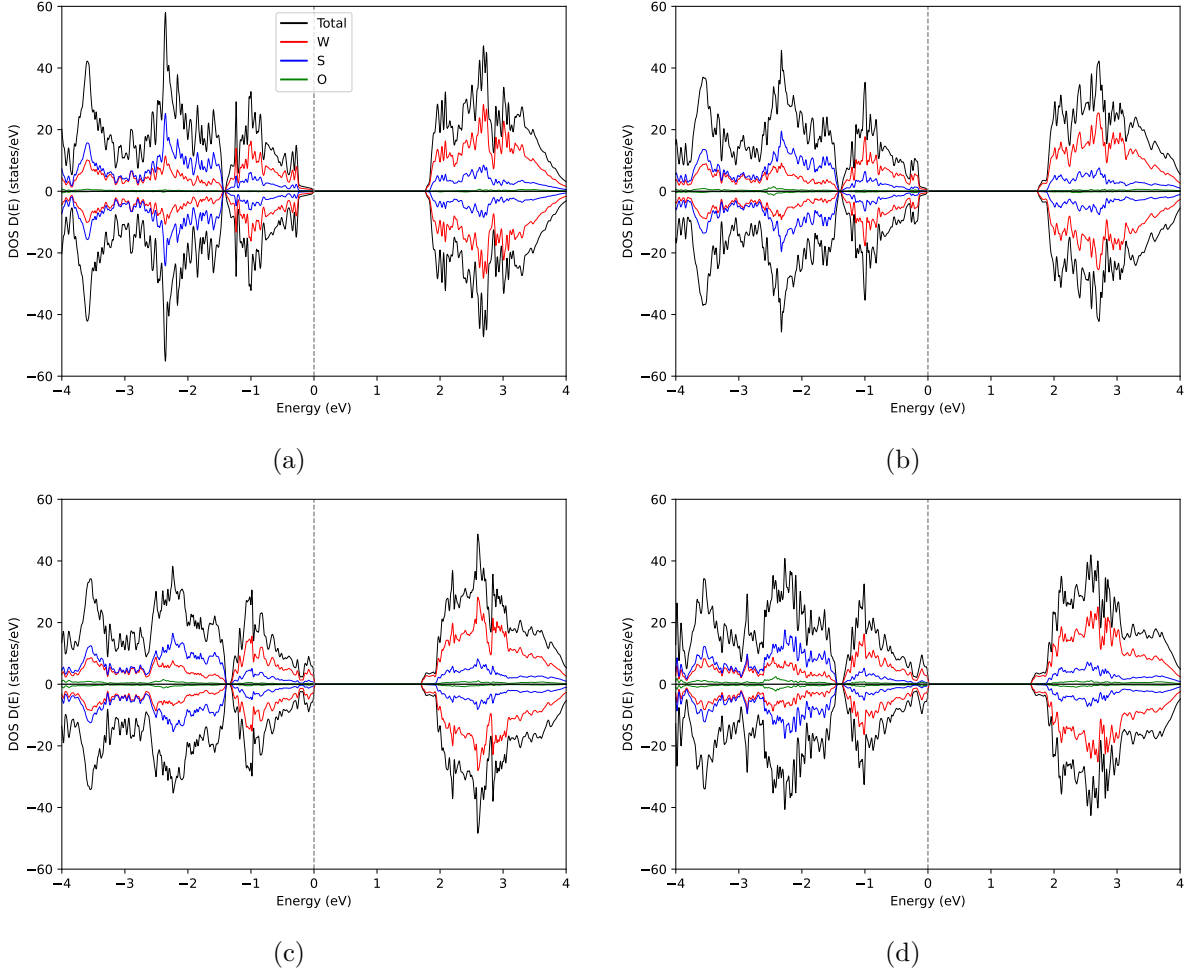


Figure 3.3: The total and partial density of states (DOS and PDOS) are shown for oxygen-doped WS₂ systems with doping concentrations of (a) 3%, (b) 6%, (c) 9%, and (d) 12%. In each plot, the dotted line marks the Fermi level.

Energy	Pristine	Sulfur-Vacancy				Oxygen-Doping			
		3%	6%	9%	12%	3%	6%	9%	12%
E_g	1.861	1.400	1.295	1.110	0.955	1.770	1.74	1.705	1.630

Table 3.1: Calculated Bandgap of the system for different Sulfur vacancy and Oxygen doping in WS₂ monolayers (energy in eV).

Chapter 4

CONCLUSION

4.1 Conclusion

In this study, the electronic properties of pristine, sulfur-vacant & oxygen-doped WS₂ monolayers were systematically analyzed using first-principles calculations. The pristine WS₂ monolayer exhibits a clear semiconducting nature with a Direct-band gap of around 1.86 eV and no spin polarization, confirming its non-magnetic ground state. The electronic structure near the band edges is primarily governed by tungsten d orbitals with contributions from sulfur p states.

The introduction of sulfur vacancies significantly alters the electronic structure by generating defect-induced states near the Fermi-level. These states reduces band gap and, at higher vacancy concentrations. Despite the presence of unsaturated bonds, no spin polarization is observed, indicating that sulfur vacancies alone are insufficient to induce magnetism in WS₂ within the studied configurations.

In contrast, oxygen substitution at sulfur vacancy sites effectively modifies the defect states by passivating the dangling bonds associated with missing sulfur atoms. The electronic states near the Fermi-level remain dominated by tungsten d orbitals, with oxygen p states contributing primarily to the valence region. Similar to the pristine and vacancy systems, no magnetic behavior is observed, as evidenced by the absence of exchange splitting in the spin-resolved density of states.

Overall, the study demonstrates that while sulfur vacancies degrade the electronic properties of WS₂ by introducing defect states, oxygen doping acts as an effective defect passivator that stabilizes the electronic structure without inducing magnetism. These findings provide valuable insights into defect engineering in transition metal dichalcogenides and suggest that controlled doping can be used to tune electronic properties for potential nanoelectronic applications.

4.2 Scope of Future work

As a continuation of this work, further investigations will focus on inducing magnetism in WS_2 through alternative defect engineering strategies. In particular, the creation of tungsten vacancies (V_{W}) and substitutional doping with transition metal atoms at tungsten sites will be explored. Unlike sulfur vacancies, which do not generate spin polarization, tungsten-site modifications are expected to introduce localized d-electron states that can give rise to finite magnetic moments.

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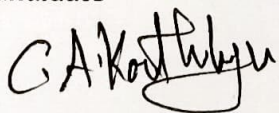
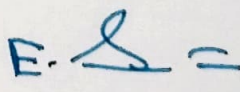
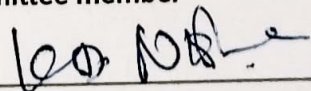
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